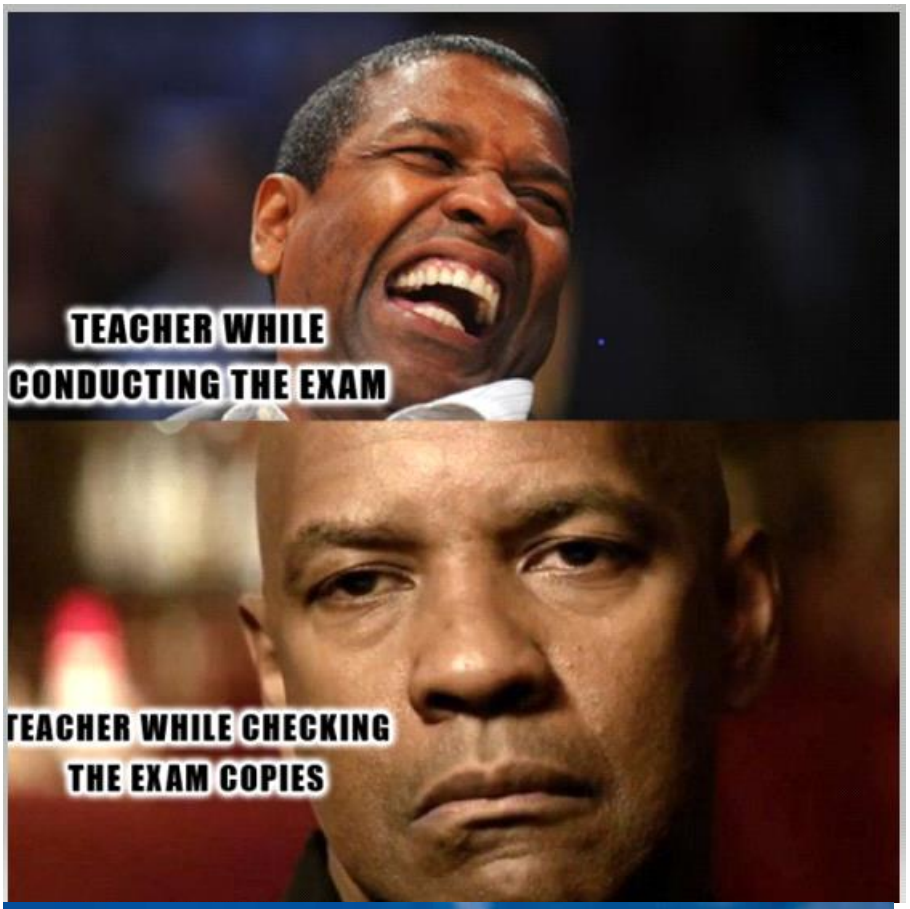
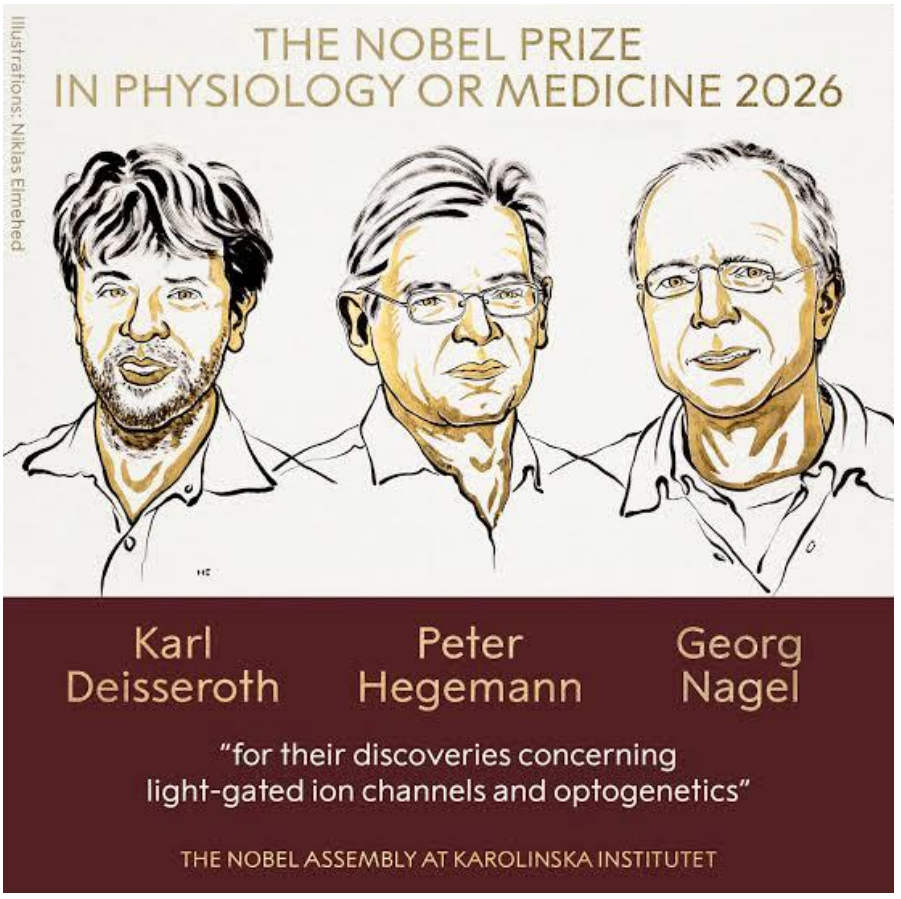
Lec-08-Week-09

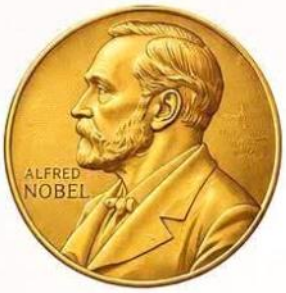
Sunday, 27 September 2026 1:32 pm






Karl Deisseroth
Howard Hughes Medical Institute
Stanford University
Stanford, USA
Peter Hegemann
Humboldt University of Berlin
Berlin, Germany
Georg Nagel
University of Würzburg
Würzburg, Germany


THE NOBEL PRIZE
IN PHYSIOLOGY OR MEDICINE 2026
Illustrations: Niklas Elmehed
Karl Deisseroth
Peter Hegemann
Georg Nagel
"for their discoveries concerning
light-gated ion channels and optogenetics"
THE NOBEL ASSEMBLY AT KAROLINSKA INSTITUTET



NOBEL PRIZE IN MEDICINE 2026

Chemical World

AWARDED FOR DISCOVERIES CONCERNING

LIGHT-GATED ION CHANNELS AND OPTOGENETICS

LIGHT
BIOLOGY
BRAIN
HEALTH
BETTER
LIVES

"For uncovering how light can control the electrical activity of cells, enabling precise control of brain circuits and revolutionizing neuroscience and medicine."



Karl Deisseroth

USA

Neuroscientist
Stanford University

Pioneer in applying optogenetics to study and manipulate brain circuits.



Peter Hegemann

Germany

Biophysicist
Humboldt University of Berlin

Discovered light-sensitive ion channels in algae, including channelrhodopsin.



Georg Nagel

Germany

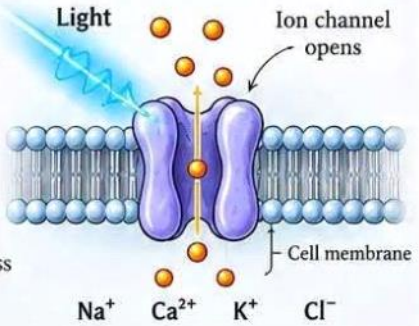
Biophysicist
University of Würzburg

Discovered another light-gated ion channel, halorhodopsin.

LIGHT-GATED ION CHANNELS

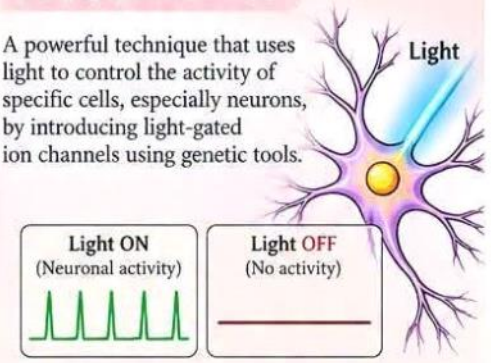
These are proteins in the cell membrane that open or close in response to light of a specific wavelength.

- Absorb light
- Change shape
- Open/close ion channel
- Allow specific ions to pass (Na^+, K^+, Ca^{2+}, Cl^-)



OPTOGENETICS

A powerful technique that uses light to control the activity of specific cells, especially neurons, by introducing light-gated ion channels using genetic tools.



MAJOR IMPACTS

Neuroscience
Understanding how brain circuits work.

Neurological Disorders
New approaches for epilepsy, Parkinson's, depression and more.

Vision Restoration
Potential therapies for blindness.

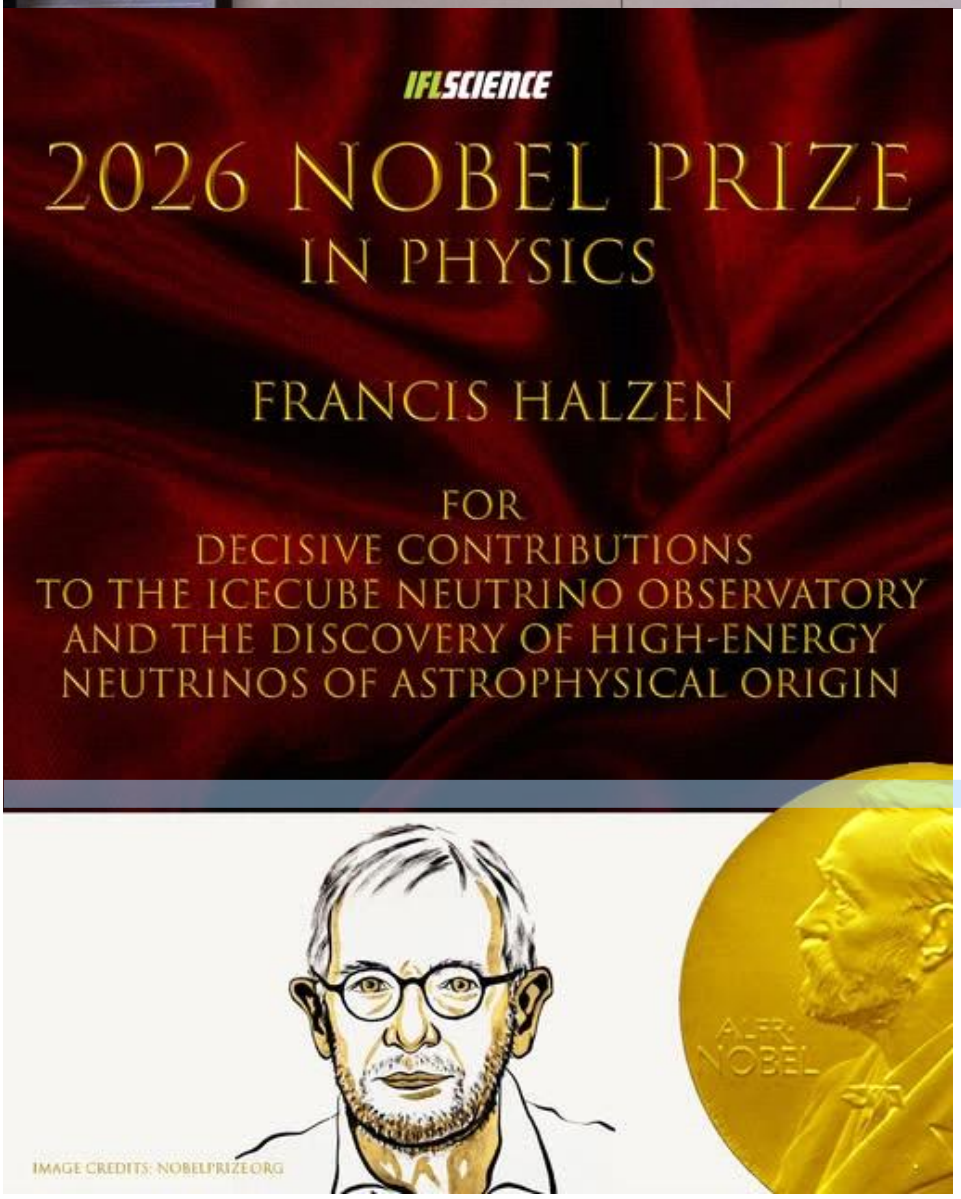
Drug Development
Better models to study brain function and diseases.

Mapping Neural Circuits
High-precision control of specific neurons in living brains.

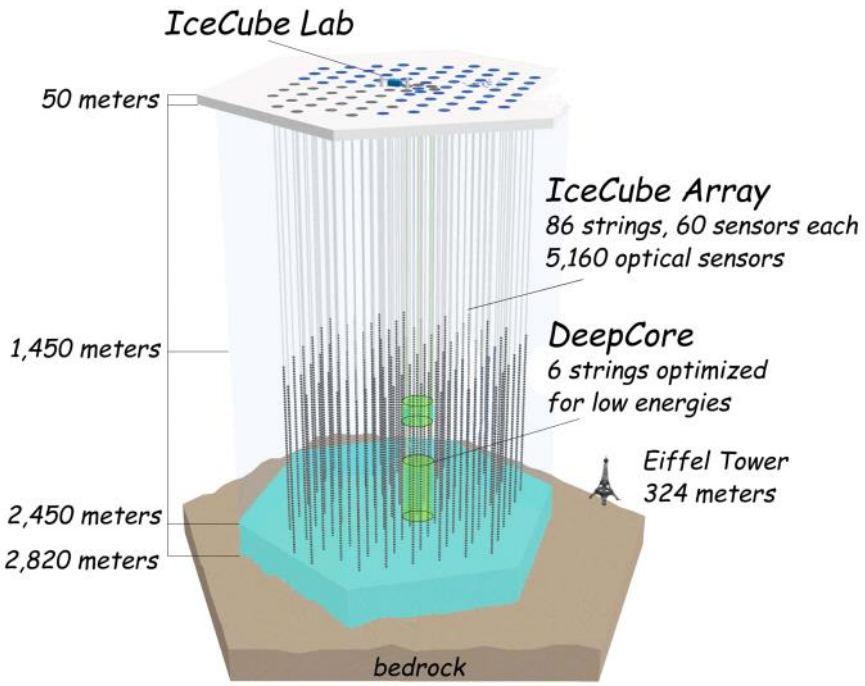
Their pioneering work has transformed neuroscience and opened new possibilities for treating a wide range of diseases.

Chemical World

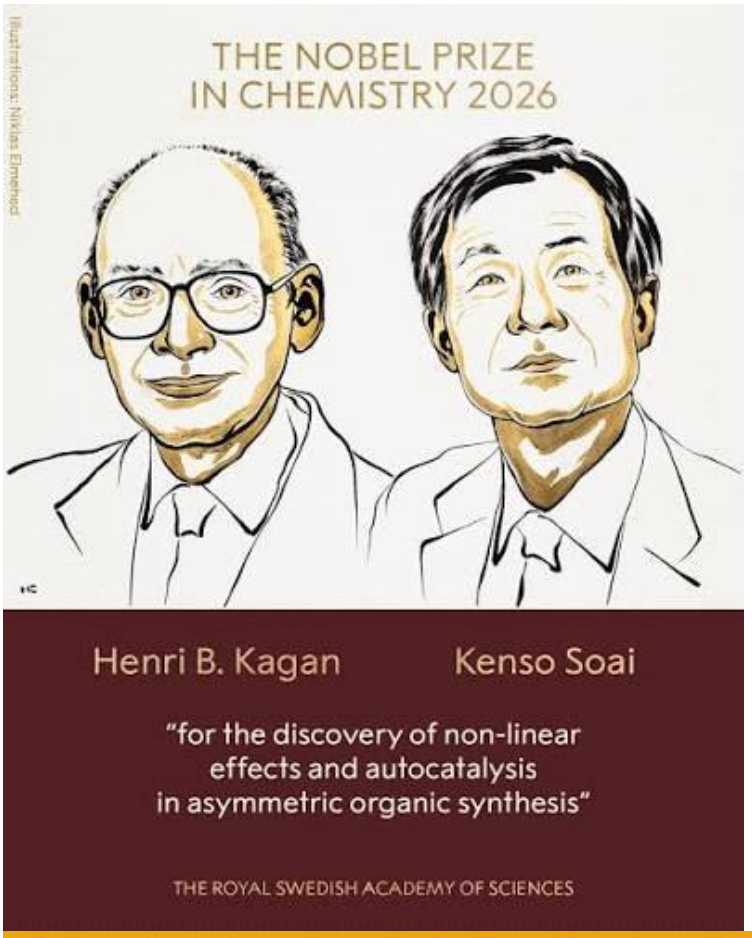

NOBELPRISET I FYSIK 2026
THE NOBEL PRIZE IN PHYSICS 2026
KUNGL. VETENSKAPS AKADEMIEN
"för avgörande bidrag till IceCube-observatoriet och upptäckten av högenergetiska neutriner av astrofysikaliskt ursprung"
"for decisive contributions to the IceCube Neutrino Observatory and the discovery of high-energy neutrinos of astrophysical origin"
Francis Halzen
University of Wisconsin–Madison, USA
Born 1944, Belgium
KUNGL. VETENSKAPS AKADEMIEN

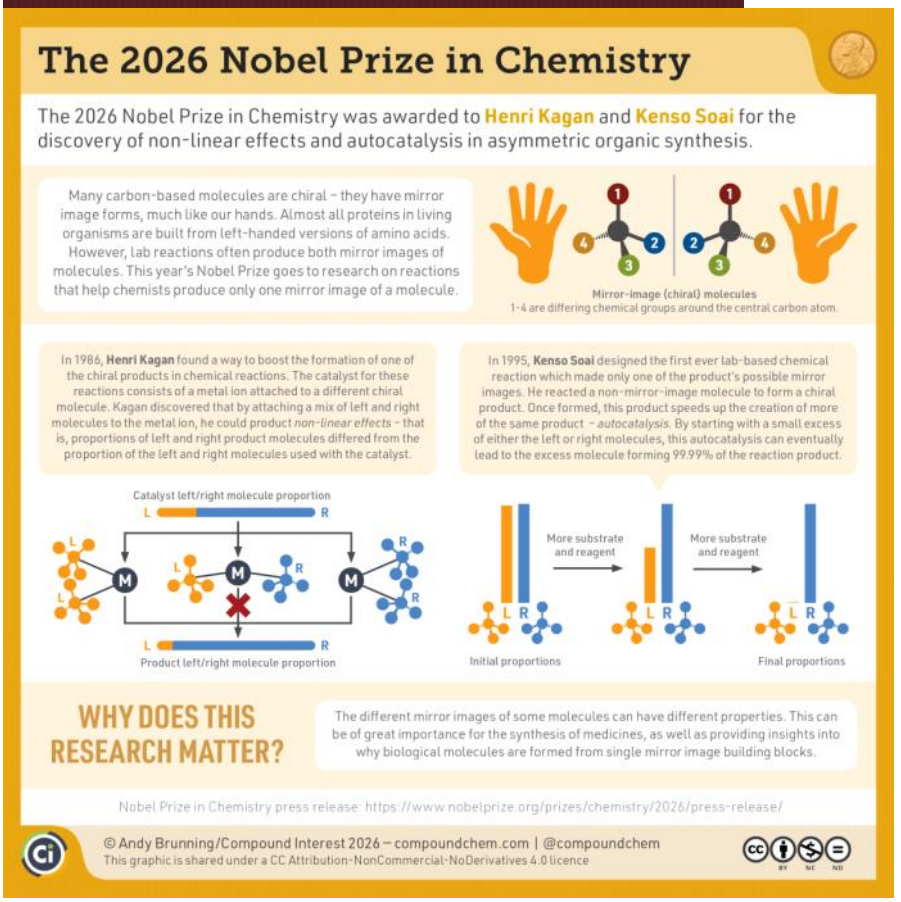

IFLSCIENCE
2026 NOBEL PRIZE
IN PHYSICS
FRANCIS HALZEN
FOR
DECISIVE CONTRIBUTIONS
TO THE ICECUBE NEUTRINO OBSERVATORY
AND THE DISCOVERY OF HIGH-ENERGY
NEUTRINOS OF ASTROPHYSICAL ORIGIN
IMAGE CREDITS: NOBELPRIZE.ORG




IceCube Lab
50 meters
IceCube Array
86 strings, 60 sensors each
5,160 optical sensors
1,450 meters
DeepCore
6 strings optimized
for low energies
Eiffel Tower
324 meters
2,450 meters
2,820 meters
bedrock


NOBELPRISET I KEMI 2026
THE NOBEL PRIZE IN CHEMISTRY 2026
AKADEMIEN
"för upptäckt av icke-linjära effekter och autokatalys i asymmetrisk organisk syntes"
"for the discovery of non-linear effects and autocatalysis in asymmetric organic synthesis"
Henri Kagan
Université Paris-Sud, France
Born 1930, France
Kenso Soai
Tokyo University of Science, Japan
Born 1950, Japan


Illustrations: Niklas Elmehed
THE NOBEL PRIZE
IN CHEMISTRY 2026
Henri B. Kagan
Kenso Soai
"for the discovery of non-linear effects and autocatalysis in asymmetric organic synthesis"
THE ROYAL SWEDISH ACADEMY OF SCIENCES


The 2026 Nobel Prize in Chemistry
The 2026 Nobel Prize in Chemistry was awarded to Henri Kagan and Kenso Soai for the discovery of non-linear effects and autocatalysis in asymmetric organic synthesis.
Many carbon-based molecules are chiral – they have mirror image forms, much like our hands. Almost all proteins in living organisms are built from left-handed versions of amino acids. However, lab reactions often produce both mirror images of molecules. This year's Nobel Prize goes to research on reactions that help chemists produce only one mirror image of a molecule.
Mirror-image (chiral) molecules
1-4 are differing chemical groups around the central carbon atom.
In 1986, Henri Kagan found a way to boost the formation of one of the chiral products in chemical reactions. The catalyst for these reactions consists of a metal ion attached to a different chiral molecule. Kagan discovered that by attaching a mix of left and right molecules to the metal ion, he could product non-linear effects – that is, proportions of left and right product molecules differed from the proportion of the left and right molecules used with the catalyst.
In 1995, Kenso Soai designed the first ever lab-based chemical reaction which made only one of the product's possible mirror images. He reacted a non-mirror-image molecule to form a chiral product. Once formed, this product speeds up the creation of more of the same product – autocatalysis. By starting with a small excess of either the left or right molecules, this autocatalysis can eventually lead to the excess molecule forming 99.99% of the reaction product.
Catalyst left/right molecule proportion
Product left/right molecule proportion
More substrate and reagent
More substrate and reagent
Initial proportions
Final proportions
WHY DOES THIS RESEARCH MATTER?
The different mirror images of some molecules can have different properties. This can be of great importance for the synthesis of medicines, as well as providing insights into why biological molecules are formed from single mirror image building blocks.
Nobel Prize in Chemistry press release: https://www.nobelprize.org/prizes/chemistry/2026/press-release/
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BRACE
YOURSELVES
FOURIER SERIES
ARE COMING...



"There cannot be a language more universal and more simple, more free from errors and obscurities... more worthy to express the invariable relations of all natural things [than mathematics]." – Jean-Baptiste Joseph Fourier

A brilliant 18th century French mathematician, Jean-Baptiste Joseph Fourier is remembered for his studies on heat conduction, the discovery of the Fourier series, and his aid in the finding of the greenhouse effect. Today, let's take a look at his life and many accomplishments...

Jean-Baptiste Joseph Fourier was born on March 21, 1768, in Auxerre, France. Orphaned by age 10, Fourier's childhood was difficult. He found escape in his studies at the École Royale Militaire, a military school in Auxerre where students were taught the importance of science and math. By age 13, Fourier had grown a strong appreciation for both subjects, especially the latter, staying up late studying equations. In fact, it is said that Fourier collected discarded candles to use as a light source when working in the dark.

Fourier Series

Fourier series are infinite series that represent periodic functions in terms of cosines and sines. As such, Fourier series are of greatest importance to the engineer and applied mathematician. To define Fourier series, we first need some background material. A function $f(x)$ is called a **periodic function** if $f(x)$ is defined for all real x, except

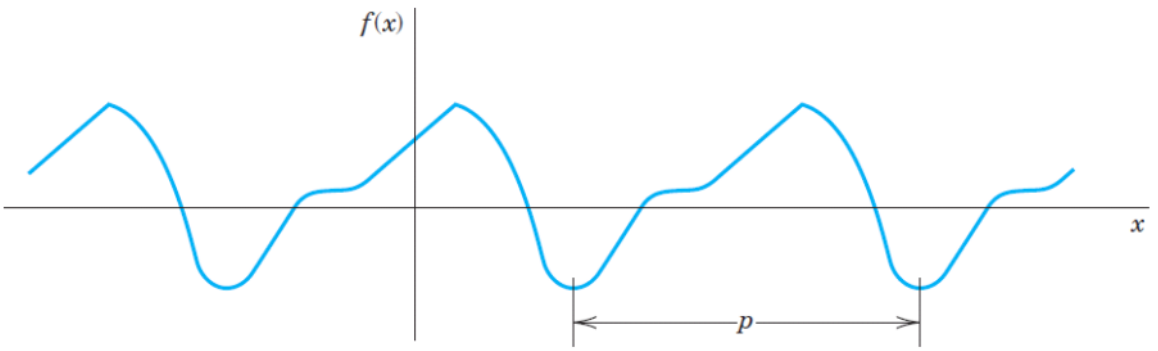
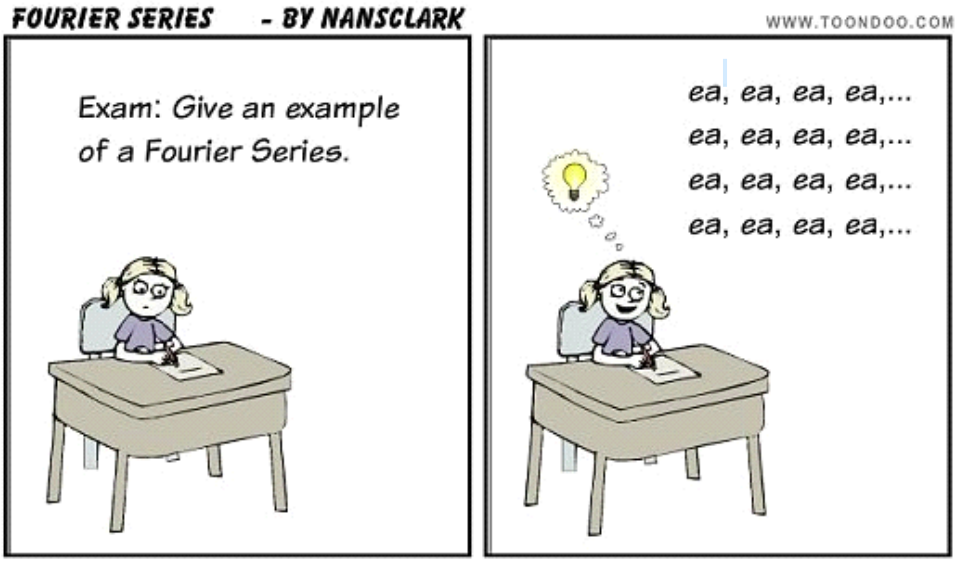
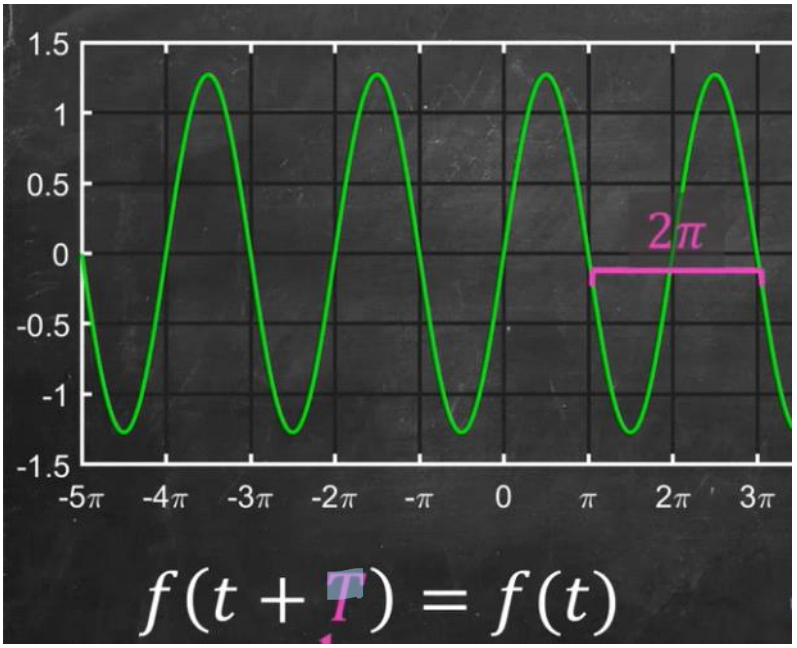


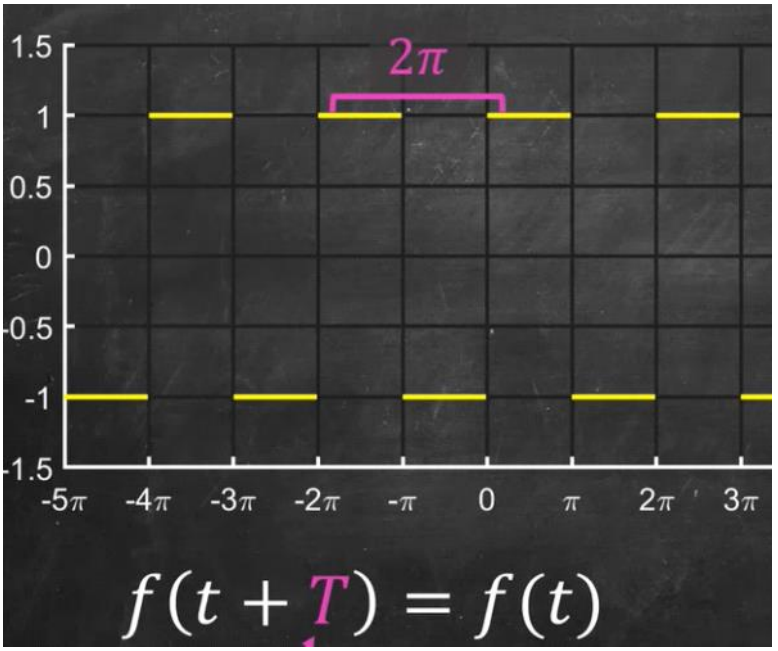
Fig. 258. Periodic function of period p



possibly at some points, and if there is some positive number p, called a **period** of $f(x)$, such that

(1) $$f(x + p) = f(x)$$ for all x.





Continuous approximation of discontinuous approximation

The smallest positive period is often called the *fundamental period.* (See Probs. 2–4.)

Familiar periodic functions are the cosine, sine, tangent, and cotangent. Examples of functions that are not periodic are x, x^2, x^3, e^x, $\cosh x$, and $\ln x$, to mention just a few.

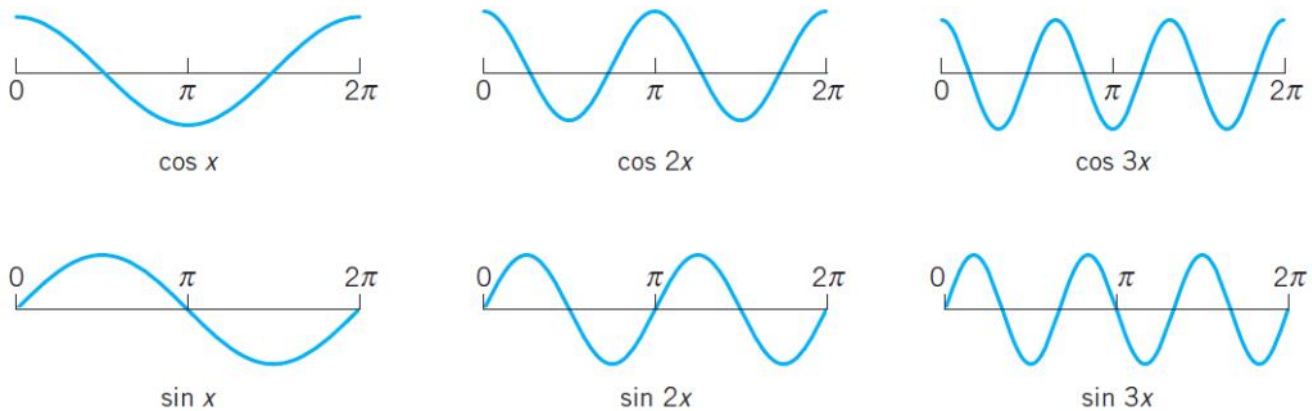
If $f(x)$ has period p, it also has the period $2p$ because (1) implies $f(x + 2p) = f([x + p] + p) = f(x + p) = f(x)$, etc.; thus for any integer $n = 1, 2, 3, \cdots$,

(2) $$f(x + np) = f(x)$$ for all x.

Furthermore if $f(x)$ and $g(x)$ have period p, then $af(x) + bg(x)$ with any constants a and b also has the period p.

(3) $$1, \quad \cos x, \quad \sin x, \quad \cos 2x, \quad \sin 2x, \cdots, \quad \cos nx, \quad \sin nx, \cdots.$$

All these functions have the period 2π. They form the so-called **trigonometric system**. Figure 259 shows the first few of them (except for the constant 1, which is periodic with any period).

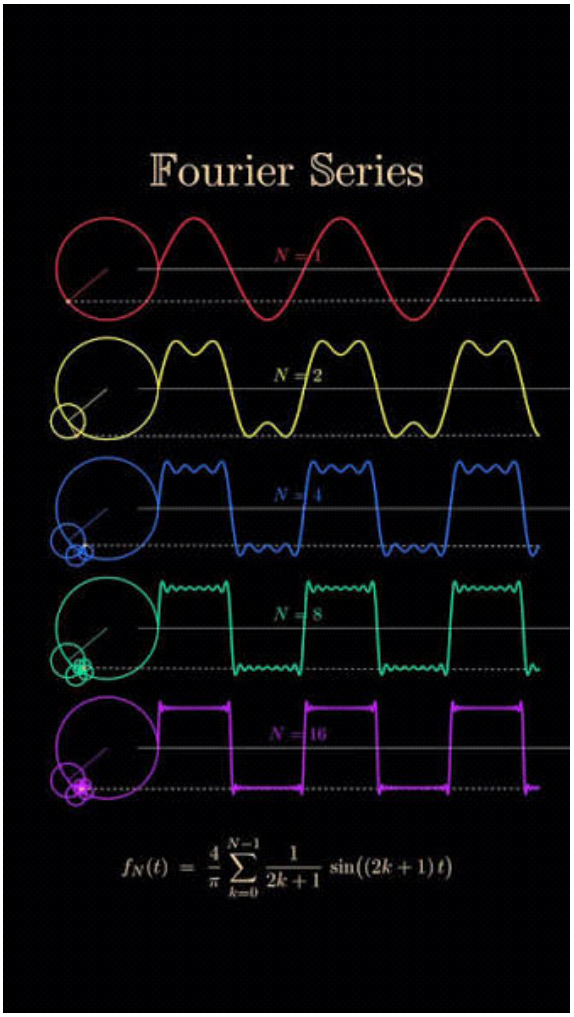


The series to be obtained will be a **trigonometric series**, that is, a series of the form

$$a_0 + a_1 \cos x + b_1 \sin x + a_2 \cos 2x + b_2 \sin 2x + \cdots$$

$$(4) \qquad = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx).$$

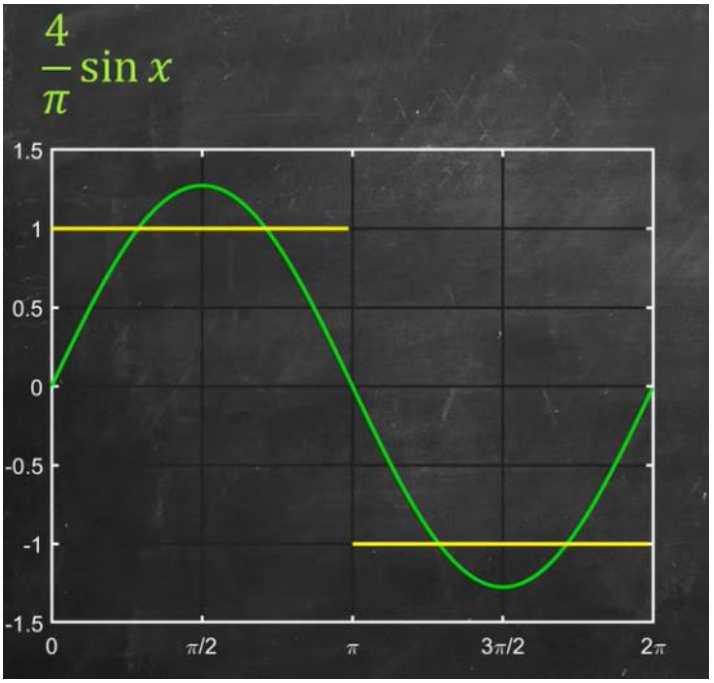
$a_0, a_1, b_1, a_2, b_2, \cdots$ are constants, called the **coefficients** of the series. We see that each term has the period 2π. Hence *if the coefficients are such that the series converges, its sum will be a function of period* 2π.

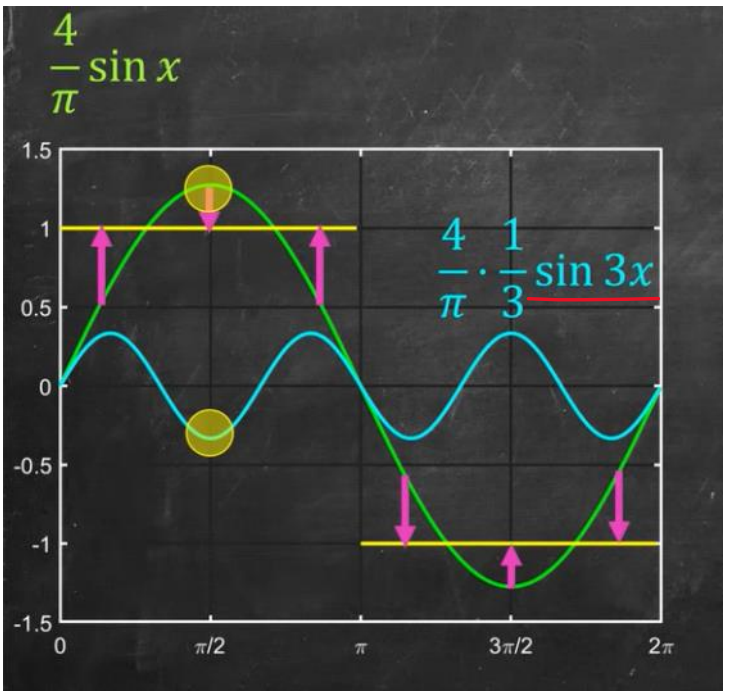


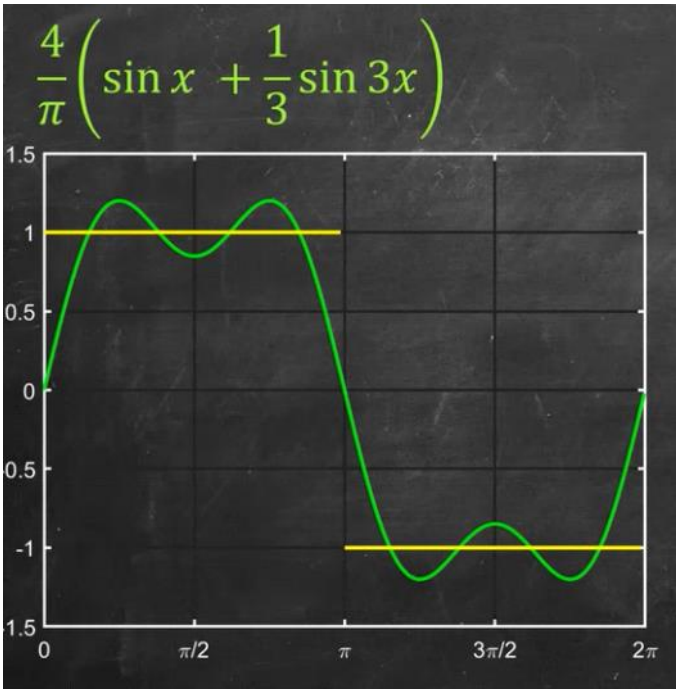
Fourier series of square wave

Fourier series expansion

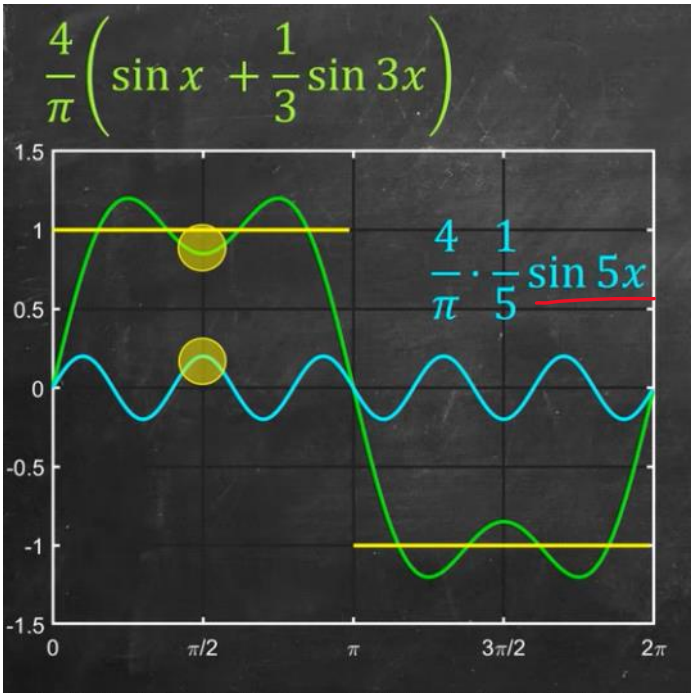
Visualizing Fourier Series


$\frac{4}{\pi}\sin x$
1.5
1
0.5
0
-0.5
-1
-1.5
0
π/2
π
3π/2
2π

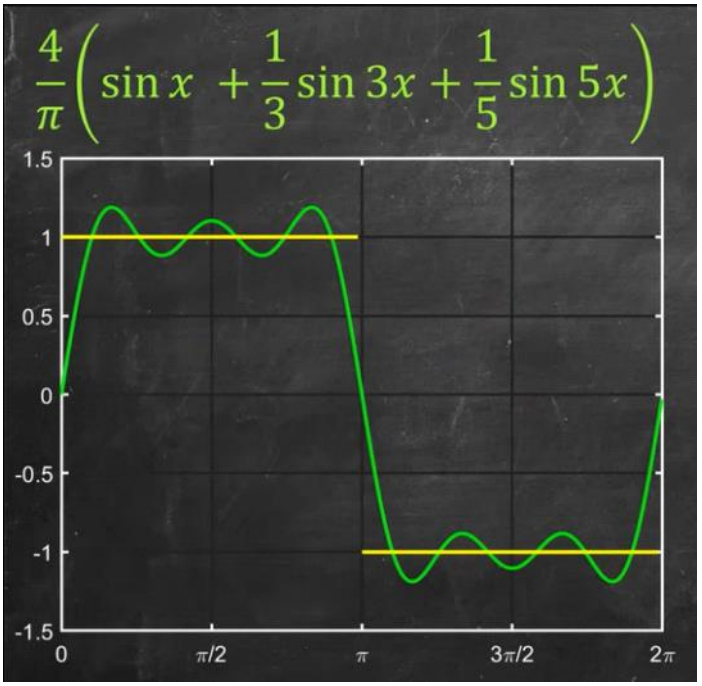

$\frac{4}{\pi}\sin x$
$\frac{4}{\pi}\cdot\frac{1}{3}\sin 3x$
1.5
1
0.5
0
-0.5
-1
-1.5
0
π/2
π
3π/2
2π

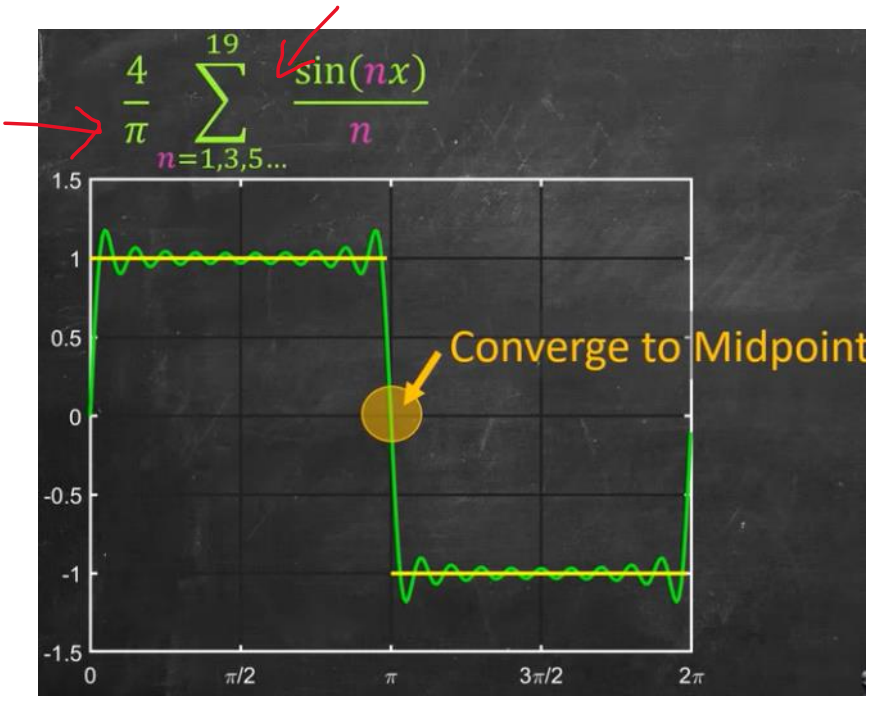


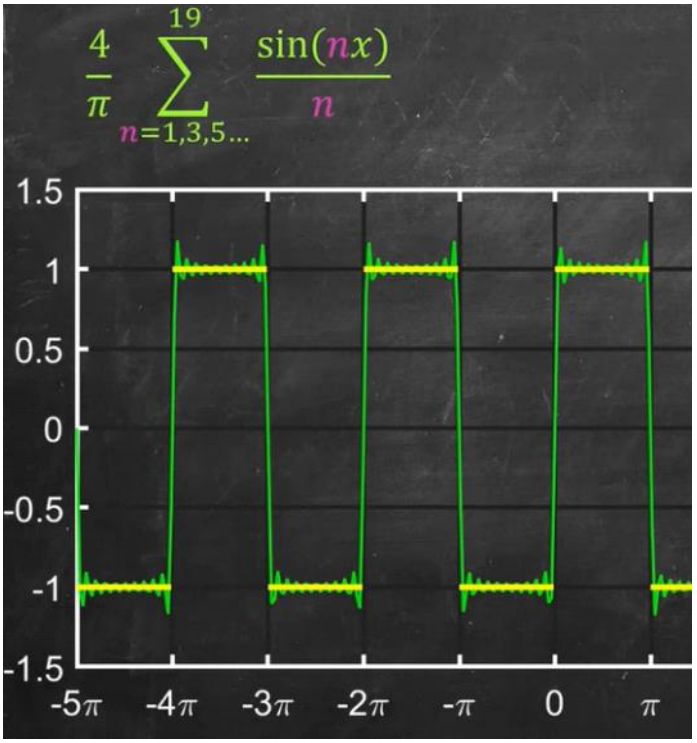
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$\frac{4}{\pi}\left(\sin x + \frac{1}{3}\sin 3x + \frac{1}{5}\sin 5x\right)$
1.5
1
0.5
0
-0.5
-1
-1.5
0
π/2
π
3π/2
2π


$\frac{4}{\pi}\sum_{n=1,3,5\ldots}^{19}\frac{\sin(nx)}{n}$
Converge to Midpoint
1.5
1
0.5
0
-0.5
-1
-1.5
0
π/2
π
3π/2
2π



$$f(x) = a_0 + \sum_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx)$$

If $f(t)$ has period 2π,

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nt + \sum_{n=1}^{\infty} b_n \sin nt$$

Ex: Square Wave:

$$\frac{4}{\pi}\sum_{n=1,3,5\ldots}^{19}\frac{\sin(nx)}{n}$$

$$a_n = 0 \qquad b_n = \begin{cases} \frac{4}{\pi}\cdot\frac{1}{n}, & n \text{ odd} \\ 0, & n \text{ even} \end{cases}$$

If $f(t)$ defined on, $[-L, L)$

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{\pi}{L}nt\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{\pi}{L}nt\right)$$

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{\pi}{L} nt\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{\pi}{L} nt\right)$$

Future Questions:

1) How do we find the a_n and b_n?

2) When do Fourier Series Converge?

3) How does this help Differential equations?

Exercises: Useful Integrals

1) $\int_{-\pi}^{\pi} \sin mx\, dx = 0.$

2) $\int_{-\pi}^{\pi} \cos mx\, dx = 0.$

3) $\int_{-\pi}^{\pi} \sin^2 mx\, dx = \pi, \quad m \neq 0.$

4) $\int_{-\pi}^{\pi} \cos^2 mx\, dx = \pi, \quad m \neq 0.$

5) $$\int_{-\pi}^{\pi} \sin nx \cos mx\, dx = \begin{cases} 0, & n \neq m. \\ \pi, & n = m \neq 0. \end{cases}$$

6) $$\int_{-\pi}^{\pi} \cos nx \cos mx\, dx = \begin{cases} 0, & n \neq m \\ \pi, & n = m \neq 0. \end{cases}$$

7) $$\int_{-\pi}^{\pi} \sin nx \sin mx\, dx = \begin{cases} 0, & n \neq m \\ \pi, & n = m \neq 0. \end{cases}$$

THE ORTHOGONALITY RELATIONS
— Orthogonal Functions

Recall: $\sin(A+B) = \sin A\cos B + \cos A\sin B$
$\sin(A-B) = \sin A\cos B - \cos A\sin B$ } —— †

† $\Rightarrow$ $\boxed{\sin A\cos B = \frac{1}{2}\left[\sin(A+B) + \sin(A-B)\right]}$.

$\cos(A+B) = \cos A\cos B - \sin A\sin B$
$\cos(A-B) = \cos A\cos B + \sin A\sin B$ } —— ††

†† $\Rightarrow$ $\boxed{\begin{aligned}\cos A\cos B &= \frac{1}{2}\left[\cos(A+B) + \cos(A-B)\right]\\ \sin A\sin B &= \frac{1}{2}\left[\cos(A-B) - \cos(A+B)\right]\end{aligned}}$

$$\int_{-\pi}^{\pi} \sin nx\cos mx\,dx \qquad , n \neq m.$$

$$= \frac{1}{2}\int_{-\pi}^{\pi}\left[\sin(n+m)x + \sin(n-m)x\right]dx$$

$$= \frac{1}{2}\left[-\frac{1}{n+m}\cos(n+m)x - \frac{1}{(n-m)}\cos(n-m)x\right]_{-\pi}^{\pi}$$

$$= \frac{1}{2}\left\{\left[-\frac{1}{n+m}\cos(n+m)\pi - \frac{1}{(n-m)}\cos(n-m)\pi\right] - \left[-\frac{1}{n+m}\cos(n+m)\pi - \frac{1}{(n-m)}\cos(n-m)\pi\right]\right\}$$

$$= 0.$$

$$\int_{-\pi}^{\pi} \sin mx\cos mx\,dx = \frac{1}{2}\int_{-\pi}^{\pi}\sin 2mx\,dx \qquad , n = m.$$

$$= -\frac{1}{4m}\cos 2mx\Big|_{-\pi}^{\pi}$$

$$=$$

- $\boxed{\int_{-\pi}^{\pi}\sin nx\cos mx\,dx = 0 \quad , m,n \in \mathbb{N}.}$

Determination of a_0: Integrate term-by-term

$$\int_{-\pi}^{\pi} f(x)\,dx = \int_{-\pi}^{\pi}\left[\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx\right] dx$$

$$= \frac{a_0}{2}\int_{-\pi}^{\pi} dx + \sum_{n=1}^{\infty} a_n \int_{-\pi}^{\pi} \cos nx\,dx + b_n \int_{-\pi}^{\pi} \sin nx\,dx$$

$$= \frac{a_0}{2} \times 2\pi + \sum_{n=1}^{\infty} a_n \times 0 + b_n \times 0$$

$$\therefore \quad \boxed{a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx}$$

Determination of a_m:

Multiply $f(x)$ by $\cos mx$ and integrate term-by-term:

$$\int_{-\pi}^{\pi} f(x)\cos mx\,dx = \int_{-\pi}^{\pi}\left[\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx\right]\cos mx\,dx$$

$$= \int_{-\pi}^{\pi} \frac{a_0}{2}\cos mx\,dx + \sum_{n=1}^{\infty} a_n \int_{-\pi}^{\pi} \cos nx \cos mx\,dx + b_n \int_{-\pi}^{\pi} \sin nx \cos mx\,dx$$

$$= 0 + a_m \times \pi + \sum_{n=1}^{\infty} b_n \times 0$$

$$\therefore \quad \boxed{a_m = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos mx\,dx, \; m = 1, 2, \ldots.}$$

Determination of b_m:

Multiply $f(x)$ by $\sin mx$ and integrate term-by-term:

$$\int_{-\pi}^{\pi} f(x)\sin mx\,dx = \int_{-\pi}^{\pi}\left[\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx\right]\sin mx\,dx$$

$$= \int_{-\pi}^{\pi}\frac{a_0}{2}\sin mx\,dx + \sum_{n=1}^{\infty} a_n \int_{-\pi}^{\pi}\cos nx \sin mx\,dx + b_n \int_{-\pi}^{\pi}\sin nx \sin mx\,dx$$

$$= 0 + \sum_{n=1}^{\infty} a_n \times 0 + b_m \times \pi$$

$$\therefore \quad \boxed{b_m = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin mx\,dx \;,\; m = 1, 2, 3, \cdots.}$$

(0) $$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx$$

(a) $$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx \qquad n = 1, 2, \cdots$$

(b) $$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx \qquad n = 1, 2, \cdots.$$

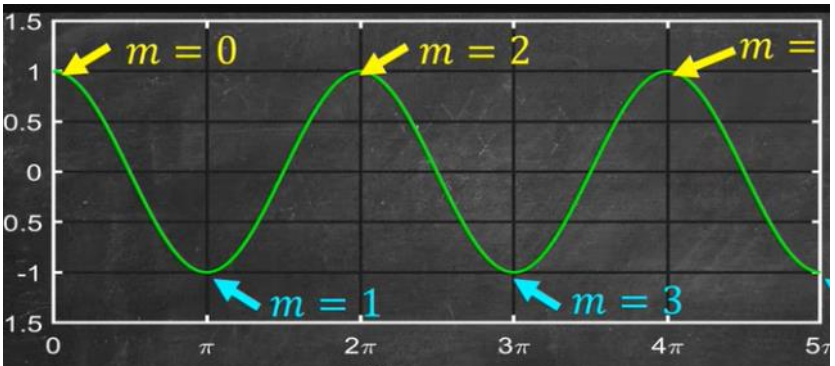
Ex: $f(t) = \begin{cases} 1, & 0 < x < \pi \\ 0, & \pi < x < 2\pi \end{cases}$

$$a_0 = 1$$

$$a_m = 0$$

$$b_m = \frac{1}{\pi}\cdot\frac{-1}{m}\cos mt\Big|_0^{\pi}$$

$$= \frac{1}{\pi}\cdot\frac{-1}{m}[\cos m\pi - 1]$$



$$b_m = \frac{1}{\pi} \cdot \frac{-1}{m} \cos mt \Big|_0^{\pi}$$

$$= \frac{1}{\pi} \cdot \frac{-1}{m} [\cos m\pi - 1]$$

$$= \frac{1}{\pi} \cdot \frac{-1}{m} [(-1)^m - 1]$$

$$= \begin{cases} 0, & m \text{ even} \\ \dfrac{2}{\pi m}, & m \text{ odd} \end{cases}$$

$$f(t) = \frac{1}{2} + \sum_{n=1,3,5\ldots}^{\infty} \frac{2}{\pi n} \sin nt$$

For f, f' piecewise continuous on $[-L, L)$,

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi t}{L} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi t}{L}$$

Converges to $f(x)$ when continuous
And to midpoint at discontinuities.

EXAMPLE 1 **Periodic Rectangular Wave (Fig. 260)**

Find the Fourier coefficients of the periodic function $f(x)$ in Fig. 260. The formula is

$$(7) \qquad f(x) = \begin{cases} -k & \text{if } -\pi < x < 0 \\ k & \text{if } 0 < x < \pi \end{cases} \quad \text{and} \quad f(x + 2\pi) = f(x).$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx = \frac{1}{\pi}\left[\int_{-\pi}^{0}(-k)\cos nx\,dx + \int_{0}^{\pi} k\cos nx\,dx\right]$$

$$= \frac{1}{\pi}\left[-k\frac{\sin nx}{n}\Big|_{-\pi}^{0} + k\frac{\sin nx}{n}\Big|_{0}^{\pi}\right] = 0$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx = \frac{1}{\pi}\left[\int_{-\pi}^{0}(-k)\sin nx\,dx + \int_{0}^{\pi} k\sin nx\,dx\right]$$

$$= \frac{1}{\pi}\left[k\frac{\cos nx}{n}\Big|_{-\pi}^{0} - k\frac{\cos nx}{n}\Big|_{0}^{\pi}\right].$$

Since $\cos(-\alpha) = \cos\alpha$ and $\cos 0 = 1$, this yields

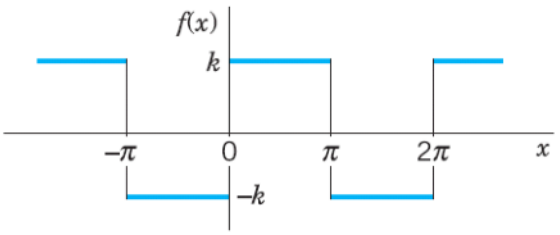
$$b_n = \frac{k}{n\pi}[\cos 0 - \cos(-n\pi) - \cos n\pi + \cos 0] = \frac{2k}{n\pi}(1 - \cos n\pi).$$

Now, $\cos\pi = -1$, $\cos 2\pi = 1$, $\cos 3\pi = -1$, etc.; in general,

$$\cos n\pi = \begin{cases} -1 & \text{for odd } n, \\ 1 & \text{for even } n, \end{cases} \qquad \text{and thus} \qquad 1 - \cos n\pi = \begin{cases} 2 & \text{for odd } n, \\ 0 & \text{for even } n. \end{cases}$$

Hence the Fourier coefficients b_n of our function are

$$b_1 = \frac{4k}{\pi}, \qquad b_2 = 0, \qquad b_3 = \frac{4k}{3\pi}, \qquad b_4 = 0, \qquad b_5 = \frac{4k}{5\pi}, \cdots.$$

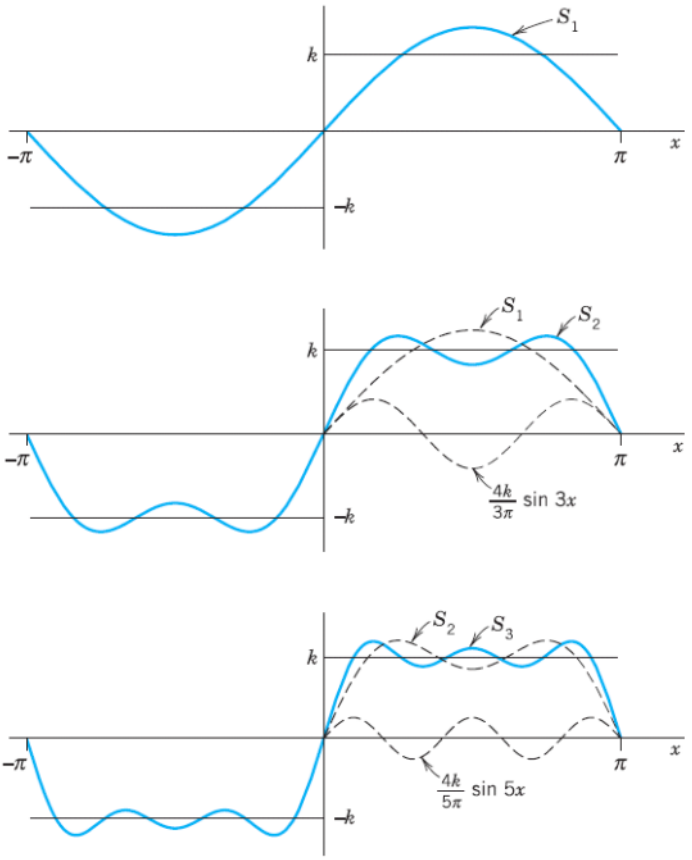


Since the a_n are zero, the Fourier series of $f(x)$ is

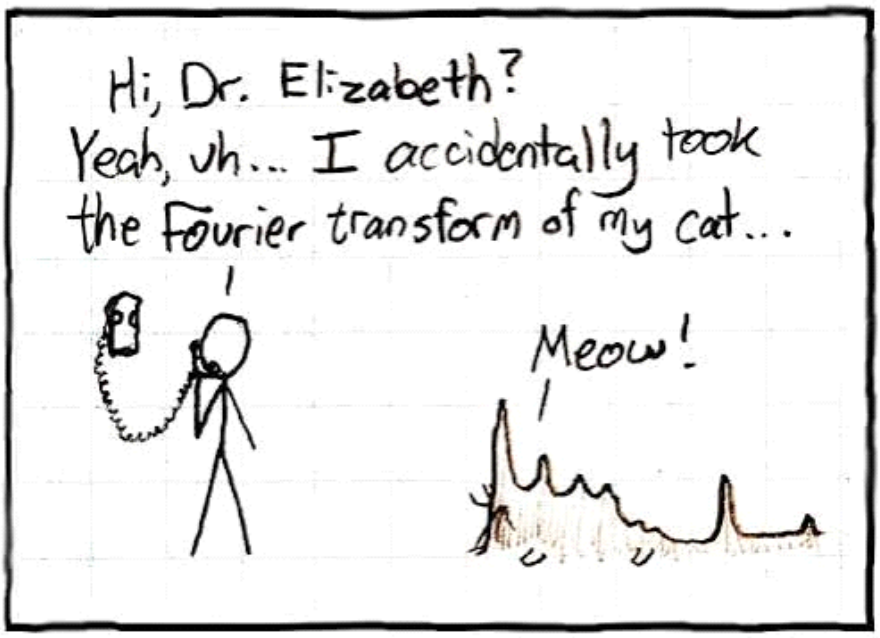
(8) $$\frac{4k}{\pi}(\sin x + \tfrac{1}{3}\sin 3x + \tfrac{1}{5}\sin 5x + \cdots).$$

The partial sums are

$$S_1 = \frac{4k}{\pi}\sin x, \qquad S_2 = \frac{4k}{\pi}\left(\sin x + \frac{1}{3}\sin 3x\right). \qquad \text{etc.}$$



ig. 261. First three partial sums of the corresponding Fourier series



THEOREM 1 **Orthogonality of the Trigonometric System (3)**

The trigonometric system (3) *is orthogonal on the interval* $-\pi \leqq x \leqq \pi$ (hence also on $0 \leqq x \leqq 2\pi$ or any other interval of length 2π because of periodicity); *that is, the integral of the product of any two functions in* (3) *over that interval is* 0, *so that for any integers n and m,*

$$\text{(9)} \qquad \text{(a)} \quad \int_{-\pi}^{\pi} \cos nx \cos mx \, dx = 0 \qquad (n \neq m)$$

$$\text{(b)} \quad \int_{-\pi}^{\pi} \sin nx \sin mx \, dx = 0 \qquad (n \neq m)$$

$$\text{(c)} \quad \int_{-\pi}^{\pi} \sin nx \cos mx \, dx = 0 \qquad (n \neq m \text{ or } n = m).$$

THEOREM 2 **Representation by a Fourier Series**

Let $f(x)$ *be periodic with period* 2π *and piecewise continuous* (see Sec. 6.1) *in the interval* $-\pi \leqq x \leqq \pi$. *Furthermore, let* $f(x)$ *have a left-hand derivative and a right-hand derivative at each point of that interval. Then the Fourier series* (5) *of* $f(x)$ [with coefficients (6)] *converges. Its sum is* $f(x)$, *except at points* x_0 *where* $f(x)$ *is discontinuous. There the sum of the series is the average of the left- and right-hand limits*[2] *of* $f(x)$ *at* x_0.

1. From Period 2π to Any Period $p = 2L$

Clearly, periodic functions in applications may have any period, not just 2π as in the last section (chosen to have simple formulas). The notation $p = 2L$ for the period is practical because L will be a length of a violin string in Sec. 12.2, of a rod in heat conduction in Sec. 12.5, and so on.

The transition from period 2π to be period $p = 2L$ is effected by a suitable change of scale, as follows. Let $f(x)$ have period $p = 2L$. Then we can introduce a new variable v such that $f(x)$, as a function of v, has period 2π. If we set

$$\text{(1)} \qquad \text{(a)} \quad x = \frac{p}{2\pi} v, \qquad \text{so that} \qquad \text{(b)} \quad v = \frac{2\pi}{p} x = \frac{\pi}{L} x$$

then $v = \pm\pi$ corresponds to $x = \pm L$. This means that f, as a function of v, has period 2π and, therefore, a Fourier series of the form

$$\text{(2)} \qquad f(x) = f\left(\frac{L}{\pi} v\right) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nv + b_n \sin nv)$$

with coefficients obtained from (6) in the last section

$$\text{(3)} \qquad a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f\left(\frac{L}{\pi} v\right) dv, \qquad a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f\left(\frac{L}{\pi} v\right) \cos nv \, dv,$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f\left(\frac{L}{\pi} v\right) \sin nv \, dv.$$

(4) $$v = \frac{\pi}{L}x, \qquad \text{we have} \qquad dv = \frac{\pi}{L}\,dx$$

and we integrate over x from $-L$ to L. Consequently, we obtain for a function $f(x)$ of period $2L$ the Fourier series

(5) $$f(x) = a_0 + \sum_{n=1}^{\infty}\left(a_n \cos\frac{n\pi}{L}x + b_n \sin\frac{n\pi}{L}x\right)$$

with the **Fourier coefficients** of $f(x)$ given by the **Euler formulas** (π/L in dx cancels $1/\pi$ in (3))

(0) $$a_0 = \frac{1}{2L}\int_{-L}^{L} f(x)\,dx$$

(6)

(a) $$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L}\,dx \qquad n = 1, 2, \cdots$$

(b) $$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\frac{n\pi x}{L}\,dx \qquad n = 1, 2, \cdots.$$

EXAMPLE 1 **Periodic Rectangular Wave**

Find the Fourier series of the function (Fig. 263)

$$f(x) = \begin{cases} 0 & \text{if} \quad -2 < x < -1 \\ k & \text{if} \quad -1 < x < \ \ 1 \\ 0 & \text{if} \quad \ \ 1 < x < \ \ 2 \end{cases} \qquad p = 2L = 4, \quad L = 2.$$

Solution. From (6.0) we obtain $a_0 = k/2$ (verify!). From (6a) we obtain

$$a_n = \frac{1}{2}\int_{-2}^{2} f(x)\cos\frac{n\pi x}{2}\,dx = \frac{1}{2}\int_{-1}^{1} k\cos\frac{n\pi x}{2}\,dx = \frac{2k}{n\pi}\sin\frac{n\pi}{2}.$$

Thus $a_n = 0$ if n is even and

$$a_n = 2k/n\pi \quad \text{if} \quad n = 1, 5, 9, \cdots, \qquad a_n = -2k/n\pi \quad \text{if} \quad n = 3, 7, 11, \cdots.$$

From (6b) we find that $b_n = 0$ for $n = 1, 2, \cdots$. Hence the Fourier series is a **Fourier cosine series** (that is, it has no sine terms)

$$f(x) = \frac{k}{2} + \frac{2k}{\pi}\left(\cos\frac{\pi}{2}x - \frac{1}{3}\cos\frac{3\pi}{2}x + \frac{1}{5}\cos\frac{5\pi}{2}x - + \cdots\right).$$

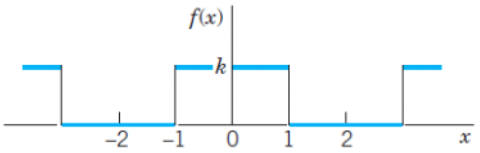


Fig. 263. Example 1

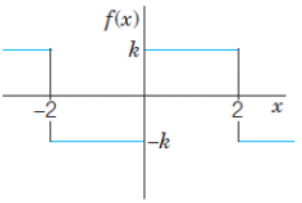


Fig. 264. Example 2

EXAMPLE 2 **Periodic Rectangular Wave. Change of Scale**

Find the Fourier series of the function (Fig. 264)

$$f(x) = \begin{cases} -k & \text{if} \quad -2 < x < 0 \\ k & \text{if} \quad 0 < x < 2 \end{cases} \qquad p = 2L = 4, \qquad L = 2.$$

Solution. Since $L = 2$, we have in (3) $v = \pi x/2$ and obtain from (8) in Sec. 11.1 with v instead of x, that is,

$$g(v) = \frac{4k}{\pi}\left(\sin v + \frac{1}{3}\sin 3v + \frac{1}{5}\sin 5v + \cdots\right)$$

the present Fourier series

$$f(x) = \frac{4k}{\pi}\left(\sin\frac{\pi}{2}x + \frac{1}{3}\sin\frac{3\pi}{2}x + \frac{1}{5}\sin\frac{5\pi}{2}x + \cdots\right).$$

Confirm this by using (6) and integrating. ■

2. Simplifications: Even and Odd Functions

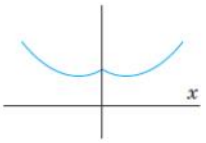


Fig. 266. Even function

If $f(x)$ is an **even function**, that is, $f(-x) = f(x)$ (see Fig. 266), its Fourier series (5) reduces to a **Fourier cosine series**

(5*)
$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi}{L}x \qquad (f \text{ even})$$

with coefficients (note: integration from 0 to L only!)

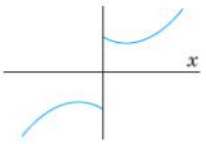


Fig. 267. Odd function

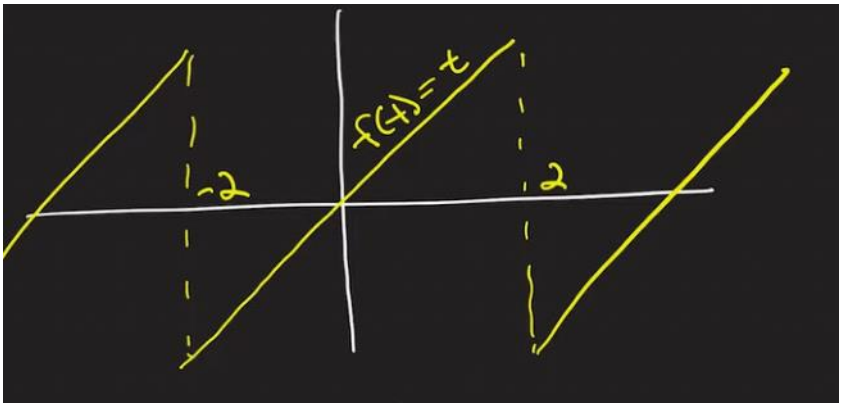
(6*)
$$a_0 = \frac{1}{L}\int_0^L f(x)\,dx, \qquad a_n = \frac{2}{L}\int_0^L f(x)\cos\frac{n\pi x}{L}\,dx, \qquad n = 1, 2, \cdots.$$

If $f(x)$ is an **odd function**, that is, $f(-x) = -f(x)$ (see Fig. 267), its Fourier series (5) reduces to a **Fourier sine series**

(5)**
$$f(x) = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi}{L}x \qquad (f \text{ odd})$$

with coefficients

(6**)
$$b_n = \frac{2}{L}\int_0^L f(x)\sin\frac{n\pi x}{L}dx.$$



$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi t}{L}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi t}{L}\right)$$

$L=2$ $\qquad$ $L=2$

$$a_n = \frac{1}{2}\int_{-2}^{2} \underbrace{t}_{f(t)\ \text{odd}} \underbrace{\cos\left(\frac{n\pi t}{2}\right)}_{\text{even}}\, dt = 0$$

$$a_0 = \frac{1}{2}\int_{-2}^{2} t \cdot 1\, dt = 0$$

$$b_n = \frac{1}{2}\int_{-2}^{2} t \sin\frac{n\pi t}{2}\, dt$$

$$b_n = \frac{2}{2}\int_{0}^{2} \underbrace{t}_{u} \underbrace{\sin\frac{n\pi t}{2}\, dt}_{dv} \qquad du = dt \qquad v = -$$

$$= -\frac{4}{n\pi}\left[\cos n\pi\right] = \frac{4}{n\pi}(-1)^{n+1}$$

$$f(t) = \sum_{n=0}^{\infty} \frac{4}{n\pi}(-1)^{n+1} \sin\frac{n\pi t}{L}$$

EXAMPLE 5 **Sawtooth Wave**

Find the Fourier series of the function (Fig. 268)

$$f(x) = x + \pi \quad \text{if} \quad -\pi < x < \pi \qquad \text{and} \qquad f(x + 2\pi) = f(x).$$

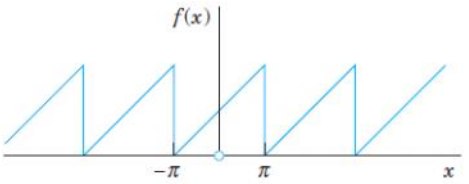


Fig. 268. The function $f(x)$. Sawtooth wave

Solution. We have $f = f_1 + f_2$, where $f_1 = x$ and $f_2 = \pi$. The Fourier coefficients of f_2 are zero, except for the first one (the constant term), which is π. Hence, by Theorem 1, the Fourier coefficients a_n, b_n are those of f_1, except for a_0, which is π. Since f_1 is odd, $a_n = 0$ for $n = 1, 2, \cdots$, and

$$b_n = \frac{2}{\pi}\int_0^{\pi} f_1(x)\sin nx\,dx = \frac{2}{\pi}\int_0^{\pi} x\sin nx\,dx.$$

Integrating by parts, we obtain

$$b_n = \frac{2}{\pi}\left[\frac{-x\cos nx}{n}\Big|_0^{\pi} + \frac{1}{n}\int_0^{\pi}\cos nx\,dx\right] = -\frac{2}{n}\cos n\pi.$$

Hence $b_1 = 2$, $b_2 = -\frac{2}{2}$, $b_3 = \frac{2}{3}$, $b_4 = -\frac{2}{4}, \cdots$, and the Fourier series of $f(x)$ is

$$f(x) = \pi + 2\left(\sin x - \frac{1}{2}\sin 2x + \frac{1}{3}\sin 3x - + \cdots\right). \qquad \text{(Fig. 269)}$$

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EXAMPLE 3 **Half-Wave Rectifier**

A sinusoidal voltage $E \sin \omega t$, where t is time, is passed through a half-wave rectifier that clips the negative portion of the wave (Fig. 265). Find the Fourier series of the resulting periodic function

$$u(t) = \begin{cases} 0 & \text{if} \quad -L < t < 0, \\ E \sin \omega t & \text{if} \quad 0 < t < L \end{cases} \qquad p = 2L = \frac{2\pi}{\omega}, \qquad L = \frac{\pi}{\omega}.$$

Solution. Since $u = 0$ when $-L < t < 0$, we obtain from (6.0), with t instead of x,

$$a_0 = \frac{\omega}{2\pi} \int_0^{\pi/\omega} E \sin \omega t \, dt = \frac{E}{\pi}$$

and from (6a), by using formula (11) in App. A3.1 with $x = \omega t$ and $y = n\omega t$,

$$a_n = \frac{\omega}{\pi} \int_0^{\pi/\omega} E \sin \omega t \cos n\omega t \, dt = \frac{\omega E}{2\pi} \int_0^{\pi/\omega} [\sin (1 + n)\, \omega t + \sin (1 - n)\, \omega t] \, dt.$$

If $n = 1$, the integral on the right is zero, and if $n = 2, 3, \cdots$, we readily obtain

$$a_n = \frac{\omega E}{2\pi} \left[-\frac{\cos (1 + n)\, \omega t}{(1 + n)\, \omega} - \frac{\cos (1 - n)\, \omega t}{(1 - n)\, \omega} \right]_0^{\pi/\omega}$$

$$= \frac{E}{2\pi} \left(\frac{-\cos (1 + n)\pi + 1}{1 + n} + \frac{-\cos (1 - n)\pi + 1}{1 - n} \right).$$

If n is odd, this is equal to zero, and for even n we have

$$a_n = \frac{E}{2\pi} \left(\frac{2}{1 + n} + \frac{2}{1 - n} \right) = -\frac{2E}{(n - 1)(n + 1)\pi} \qquad (n = 2, 4, \cdots).$$

In a similar fashion we find from (6b) that $b_1 = E/2$ and $b_n = 0$ for $n = 2, 3, \cdots$. Consequently,

$$u(t) = \frac{E}{\pi} + \frac{E}{2} \sin \omega t - \frac{2E}{\pi} \left(\frac{1}{1 \cdot 3} \cos 2\omega t + \frac{1}{3 \cdot 5} \cos 4\omega t + \cdots \right).$$

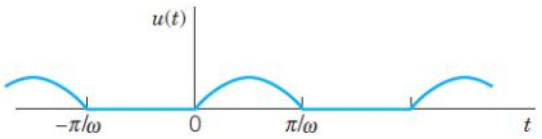


Fig. 265. Half-wave rectifier

A few useful Trigonometric Facts :

$$\sin\left(n\frac{\pi}{2}\right)=\begin{cases}0, & n \text{ even}.\\ 1, & n=1,5,9,\dots\\ -1, & n=3,7,11,\dots\end{cases}$$

$$\sin(n\pi)=0,\quad n=1,2,3,\dots.$$

$$\cos(n\pi)=\begin{cases}1, & n=2,4,6,\dots\\ -1, & n=1,3,5,\dots\end{cases},\quad \text{or}\quad \cos(n\pi)=(-1)^n,\ n=1,2,3,\dots$$

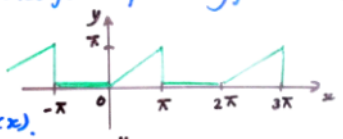
$$\cos 2\pi n=1,\quad n=1,2,3,\dots$$

$$\sin 2\pi n=0,\quad n=1,2,3,\dots$$

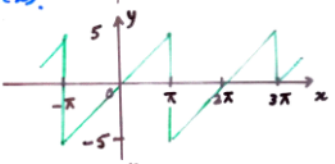
$$\cos\left(n\frac{\pi}{2}\right)=\begin{cases}1, & n=0,4,8,12,\dots\\ -1, & n=2,6,10,14,\dots\\ 0, & n=1,3,5,7,\dots\end{cases}$$

Exercises : Find the Fourier Series for the following functions :

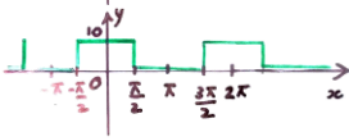
1- $f(x)=\begin{cases}0, & -\pi<x<0.\\ x, & 0<x<\pi,\end{cases}$ and $f(x+2\pi)=f(x)$.



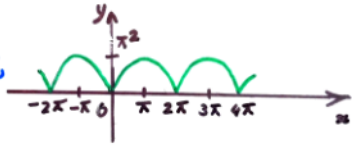
2- $f(x)=\frac{5}{\pi}x,\ -\pi<x<\pi$, and $f(x+2\pi)=f(x)$.



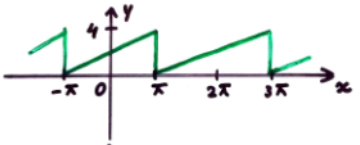
3- $f(x)=\begin{cases}0, & -\pi<x<-\frac{\pi}{2}.\\ 10, & -\frac{\pi}{2}<x<\frac{\pi}{2}.\\ 0, & \frac{\pi}{2}<x<\pi,\end{cases}$ and $f(x+2\pi)=f(x)$.



4- $f(x)=x(2\pi-x),\ 0<x<2\pi$, and $f(x+2\pi)=f(x)$.

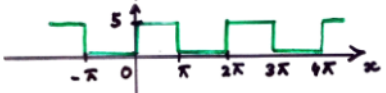


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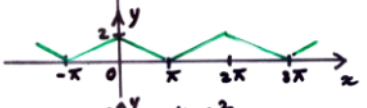
$f(x+2\pi)=f(x)$

6-



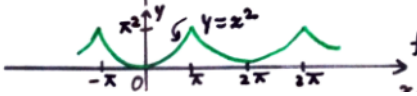
$f(x+2\pi)=f(x)$

7-



$f(x+2\pi)=f(x)$.

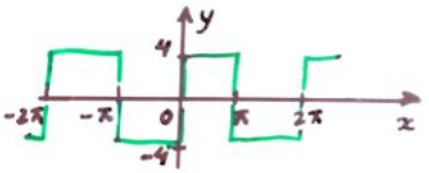
8-



$f(x+2\pi)=f(x)$.

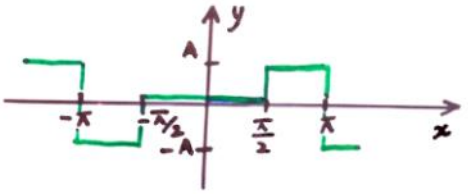
9- $f(x)=\begin{cases}-4, & -\pi<x<0\\ 4, & 0<x<\pi\end{cases}$

$f(x+2\pi)=f(x),\ \forall x.$



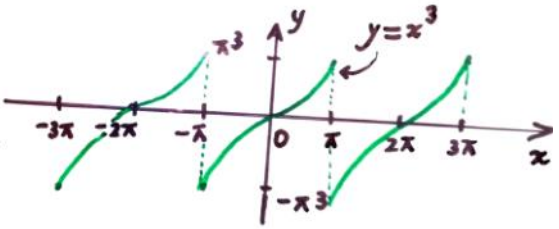
10- $f(x)=\begin{cases}-A, & -\pi<x<-\frac{\pi}{2}\\ 0, & -\frac{\pi}{2}<x<\frac{\pi}{2}\\ A, & \frac{\pi}{2}<x<\pi\end{cases}$

$f(x+2\pi)=f(x),\ \forall x.$



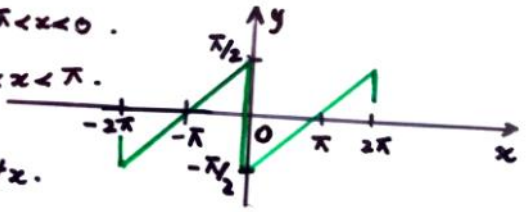
11- $f(x)=x^3,\ -\pi<x<\pi$

$f(x+2\pi)=f(x),\ \forall x.$



12- $f(x)=\begin{cases}\frac{x}{2}+\frac{\pi}{2}, & -\pi<x<0.\\ \frac{x}{2}-\frac{\pi}{2}, & 0<x<\pi.\end{cases}$

$f(x+2\pi)=f(x),\ \forall x.$

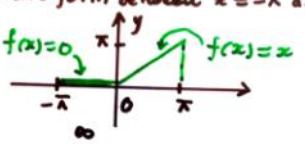


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$f(x)=\begin{cases}0, & -\pi<x<0.\\ x, & 0<x<\pi,\\ \text{and } f(x+2\pi)=f(x).\end{cases}$



Consider the wave form between $x=-\pi$ and $x=\pi$.



$$f(x)=\frac{a_0}{2}+\sum_{n=1}^{\infty}(a_n\cos nx+b_n\sin nx)$$

To find a_0:

$$a_0=\frac{1}{\pi}\int_{-\pi}^{\pi}f(x)dx=\frac{1}{\pi}\int_{-\pi}^{0}0\,dx+\frac{1}{\pi}\int_{0}^{\pi}x\,dx$$

$$=0+\frac{1}{\pi}\left[\frac{x^2}{2}\right]_0^{\pi}=\frac{\pi}{2}.$$

To find a_n:

$$a_n=\frac{1}{\pi}\int_{-\pi}^{\pi}f(x)\cos nx\,dx \qquad n=1,2,3,\ldots$$

$$=\frac{1}{\pi}\int_{-\pi}^{0}0\,dx+\frac{1}{\pi}\int_{0}^{\pi}x\cos nx\,dx$$

$$=0+\frac{1}{\pi}\left\{x\left(\frac{\sin nx}{n}\right)_0^{\pi}-\frac{1}{n}\int_0^{\pi}\sin nx\,dx\right\}$$

$$=\frac{1}{\pi n}\left[\frac{-\cos nx}{n}\right]_0^{\pi}$$

$\sin n\pi=0,\ n=1,2,3,\ldots$

$\cos n\pi=(-1)^n,\ n=1,2,3,\ldots$

$$=\frac{1}{n^2\pi}(1-\cos nx)$$

$$=\frac{1}{n^2\pi}(1-(-1)^n) \qquad n=1,2,3,\ldots$$

$\therefore\ a_n=0$, n even and $a_n=-\frac{2}{n^2\pi}$, n odd.

To find b_n:

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx \quad , n=1,2,\dots$$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} 0\sin nx\,dx + \int_{0}^{\pi} x\sin nx\,dx\right\}$$

$$= \frac{1}{\pi}\left\{0 + \left[x\left(\frac{-\cos nx}{n}\right)\right]_0^{\pi} + \frac{1}{n}\int_0^{\pi}\cos nx\,dx\right\}$$

$$= \frac{1}{\pi}\left\{-\frac{\pi}{n}\cos n\pi + \frac{1}{n^2}\sin n\pi\right\}$$

$$= -\frac{1}{n}\cos n\pi \qquad \sin n\pi = 0,\ n=1,2,3,\dots$$

$$= -\frac{1}{n}(-1)^n \quad , n=1,2,3,\dots \qquad \cos n\pi = (-1)^n,\ n=1,2,3,\dots$$

Therefore: $a_0 = \frac{\pi}{2}$; $a_n = \begin{cases} 0, & n \text{ even} \\ -\frac{2}{n^2\pi}, & n \text{ odd} \end{cases}$; $b_n = -\frac{1}{n}(-1)^n$, $n=1,2,\dots$

$$f(x) = \frac{\pi}{4} - \frac{2}{\pi}\left\{\cos x + \frac{1}{3^2}\cos 3x + \frac{1}{5^2}\cos 5x + \dots\right\} + \left\{\sin x - \frac{1}{2}\sin 2x + \frac{1}{3}\sin 3x - \frac{1}{4}\sin 4x + \dots\right\}.$$

$$\therefore a_0 = \frac{4\pi^2}{3} \ ; \ a_n = \frac{-4}{n^2} \ , \ b_n = 0 \ , \ n=1,2,\dots$$

$$f(x) = \frac{4\pi^2}{3} - 4\left\{\cos x + \frac{\cos 2x}{2^2} + \frac{\cos 3x}{3^2} + \frac{\cos 4x}{4^2} + \dots\right\}.$$

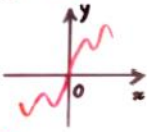
Remarks:

- Note that f is even, if $f(-x) = -f(x)$, $\forall x$.



$$\int_{-a}^{a} f(x)\,dx = 2\int_0^a f(x)\,dx \ , f \text{ even. (Why?) (Ex)}$$

- Note that f is odd, if $f(-x) = -f(x)$, $\forall x$.



$$\int_{-a}^{a} f(x)\,dx = 0 \ , f \text{ odd. (Why?) (Ex)}$$

Comments: Exercises

1) if f even, then the Fourier Coefficients $b_n = 0$, $\forall n$, and $a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)\,dx$, $a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos nx\,dx$.

2) if f odd, then the Fourier Coefficients $a_0 = 0, a_n = 0, \forall n$, and $b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\,dx$.

3) even x even = even ; odd x odd = even; and even x odd = odd.